# Homework 2 File Submission

A. Prove that if $x$ is rational and $x \neq 0$, then $\frac{1}{x}$ is rational.

Hint: Use the definition of a rational. If $x$ is rational, then it can be written as $x = \frac{p}{q}$ where p and q are non-zero integers (written $p, q \in \mathbb{Z}^*$)

Given: x is rational and $x \neq 0$.

WTP: $\frac{1}{x}$ is rational.

Proof: Since we assume x is rational, by definition it can be written in the form $x = \frac{p}{q}$, where p and $q \in \mathbb{Z}$, $q \neq 0$. Because $x \neq 0$, it follows that $p \neq 0$.
Now, let's consider the reciprocal of x: $\frac{1}{x} = \frac{1}{\frac{p}{q}} = \frac{q}{p}$.
Here, q and p are integers, and since $p \neq 0$, the denominator is also nonzero.
Therefore, $\frac{q}{p}$ is a ratio of two integers with the denominator not equal to zero.
By definition, $\frac{q}{p}$ is rational.

$\therefore$ Thus, if x is rational and $x \neq 0$, then $\frac{1}{x}$ is rational. ■

B. Prove using the notion of without loss of generality (WLOG) that $5x + 5y$ is an odd integer when $x$ and $y$ are integers of opposite parity.

Given: x and y are integers of opposite parity.

WTP: $5x + 5y$ is an odd integer.

Proof: Since x and y are integers of opposite parity, one is even and one is odd.
WLOG, assume x is even and y is odd.
1. Then there exists an integer $x = 2n$.
2. And there exists an integer, m, such that $y = 2m + 1$.

Now we can compute: $5x + 5y = 5(2n) + 5(2m+1) = 10n + 10m + 5$
$= 2(5n + 5m) + 5$

Since $5n + 5m$ is an integer, let $k = 5n + 5m$. Then, $5x + 5y = 2k + 5$.
But $2k$ is even, and adding 5 always yields an odd integer. Therefore, $5x + 5y$ is odd.
By symmetry, if instead x were odd and y were even, the proof would be the same.
Thus, our assumption of WLOG holds.

$\therefore$ Thus, if x and y are integers of opposite parity, then $5x + 5y$ is an odd integer. ■